

High-throughput synapse profiling reveals cell-type-specific spatial configurations in the fly brain

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This **important** work introduces a splitGFP-based labeling tool with an analysis pipeline for the synaptic scaffold protein bruchpilot, with tests in the adult *Drosophila* mushroom bodies, a learning center in the *Drosophila* brain. The evidence supporting the conclusions is **solid**. However, additional controls, validation of synapse-specificity, validation of activity-dependence, details on image processing, and additional functional experiments are needed to strengthen the study.

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Abstract

Characterization of intracellular synapse heterogeneity aides to understand the intricate computational logic of neuronal circuits. Despite recent advances in connectomics, the spatial patterns of synapses and their inter-individual variability remain largely unknown. Using directed split-GFP reconstitution, we achieved visualization of endogenous Bruchpilot (Brp), a presynaptic active zone (AZ) scaffold protein, in a cell-type-specific manner. By developing a high-throughput quantification pipeline, we profiled AZ structures in identified neurons of the mushroom body circuit, where intracellular synaptic patterns are crucial due to compartmentalized connectivity. Quantitative characterization of the pattern of Brp clusters across multiple individuals revealed cell-type-dependent synapse heterogeneity and stereotypy. Furthermore, we discovered previously unidentified sub-compartmental synapse configuration and its transient structural plasticity triggered by associative learning. Synapse profiling thus uncovered multi-layered organizations of AZs, ranging from neighboring synapses to consistent patterns across individuals.

Introduction

The brain is a complex network of neurons, and understanding their wiring strategies provides crucial insights into their roles in neuronal computation. Release sites in the presynaptic terminal of *Drosophila* neurons are decorated with electron-dense projections composed of the active-zone (AZ) scaffold protein Brp ^{1–7}. These Brp projections serve as a synapse marker enabling the whole-brain connectomes, comprehensive mapping of synaptic connections at the electron microscopic level ^{8–12}. While connectome-based approaches catalyzed the exploration of cell type distinctions, circuit motifs, and neurotransmitters, their application to comparisons across different conditions is still challenging due to the limited throughput of data acquisition and dense reconstruction. Since the synaptic structure and connectivity reflect substantial variability of development and experiences ^{11,13–18}, cross-individual comparisons are necessary. Fluorescence labeling offers high throughputs and contents, especially when comparing synaptic organizations across cell-types ¹⁹ or brain regions ²⁰. Nevertheless, characterizing endogenous synapses in specific cells using fluorescence-labeling remains challenging in the central nervous system (CNS) due to the dense synaptic connections from different neurons. This has driven growing interest in designing cell-type-specific fluorescence-tagging strategies for endogenous synaptic protein ²¹.

The circuit of *Drosophila* mushroom bodies (MBs) plays a central role in olfactory associative learning ²². The major intrinsic MB neuron, known as Kenyon cells (KCs) respond to odor stimulation and synapse onto the spatially segregated dendrites of MB output neurons (MBONs) that divide the MB into compartments. Distinct types of dopaminergic neurons (DANs) synapse onto specific compartments and thereby modulate KC-MBON synapses in the corresponding compartments ²³. Presynaptic calcium levels of these DANs undergo sub-compartmental GABAergic modulation and inform memory specificity, postulating the distinct synapse structures at individual release sites ²⁴. Two types of giant interneurons, the anterior paired lateral neuron (APL) and the dorsal paired medial (DPM) neuron extensively ramify across the MB lobes and provide recurrent modulations to KC synapses ^{25–32}. Considering their comprehensive projections throughout the MB lobes, the modulation of local circuits requires intracellular synaptic tuning. These studies together underscore the importance to spatially distinguish individual synapses within a cell. However, the high AZ density in the MB lobes precluded conventional approaches of fluorescent microscopy from profiling synaptic structures of specific neurons ⁹.

Here, we present a spatial analysis of individual AZs within the MB circuit at the single-cell resolution, achieved by cell-type-specific visualization of endogenous Brp using the CRISPR/Cas9-mediated split-GFP tagging system. We developed a high-throughput quantification pipeline to systematically profile Brp clusters of individual AZs in different MB-innervating neurons. Characterizing multiple structural parameters revealed a significant synaptic heterogeneity within single neurons and AZ distribution stereotypy across individuals. These cell-type-specific synapse profiles suggest that AZs are organized at multiple scales, ranging from neighboring synapses to across individuals.

Results

Establishing an experimental system for cell-type-specific profiling of active zone structures

As the molecular assembly of synapses is reported to be sensitive to the dosage of Brp, we visualized endogenous Brp instead of transgenic expression of tagged Brp ³³. To label endogenous Brp specifically in designated cell types, we employed the split-GFP tagging system ^{34,35}. Using CRISPR/Cas9, we inserted the GFP₁₁ fragment (the eleventh β -strand of the superfolder GFP) just prior to the stop codon of *brp* gene. The GFP₁₋₁₀ fragment is expressed in cells of interest by the GAL4/UAS system. The self-assembly of the split-GFP fragments is therefore directed only in GAL4-expressing cells and visualizes endogenous Brp through reconstituted GFP fluorescence (Fig. 1A). As a proof of principle, we directed GFP reconstitution and expressed a plasma membrane marker in the single serotonergic DPM neuron using *VT64246-GAL4* (Fig. 1B). Confocal microscopy revealed Brp::reconstituted GFP (Brp::rGFP) signals only to the MB lobes, aligning with DPM terminals ³². These signals were distributed as discrete puncta within plasma membrane varicosities resembling presynaptic boutons (Fig. 1B), suggesting that Brp::rGFP clusters labeled individual AZs in the DPM neuron.

To systematically profile Brp::rGFP clusters, we developed an image processing pipeline (Fig. 1A). In brief, image deconvolution was applied to reduce out-of-focus light and enhance signal sharpness. Individual Brp::rGFP clusters were then segmented using the 3D suite plugin in Fiji ³⁶, generating the 3D region-of-interests (3D ROIs) that encircle each cluster. Brp::rGFP signal intensity, volume and location of individual clusters were measured using the 3D ROIs. To optimize the segmentation, we systematically varied the parameters and calculated the F-score to measure detection accuracy against manually defined ground truth (see Materials and Methods). The F-score is near 1.0 in segmenting Brp::rGFP clusters within DPM and APL neurons and achieved > 0.90 in KCs with certain combinations of segmentation parameters (Fig. 1B).

To confirm that Brp puncta represent AZs, we quantified the numbers of Brp::rGFP clusters in DPM and APL neurons specifically and compared them to the connectome data. We found that cluster counts closely matched the number of AZs annotated by the hemibrain connectome (Fig. 1C) ⁹. This result validated our method that accurately isolated Brp::rGFP clusters corresponding to single AZs.

AZ profiles within KCs are compartmentally heterogeneous

Brp accumulation at individual AZs is known to be heterogeneous even within a single motor neuron ^{5,6,37,38}. To study such AZ heterogeneity in CNS neurons, we quantified the signal intensity of individual Brp::rGFP clusters in MB lobes. Using KC subtype-specific drivers to direct split-GFP reconstitution in γ KC (*MB009B-GAL4*), α'/β' KCs (*MB370B-GAL4*) and α/β KCs (*MB008B-GAL4*), we found that within-cluster Brp::rGFP intensities are highly diverse in all KC subtypes (Fig. 2A).

To visualize spatial patterns of Brp::rGFP intensity in each KC subtype, we reconstructed 3D distribution of Brp::rGFP clusters and color-coded them by signal intensities (Fig. 2B). We found that clusters showed noticeable intensity differences between compartments in all subtypes. Even adjacent compartments such as $\gamma 1$ and $\gamma 2$, $\beta' 2$ and $\beta' 1$, $\alpha 2$ and $\alpha 3$, showed drastic differences (Fig. 2B). Importantly, these compartmental patterns were stable across individuals (Fig. 2C) ³⁹. This result suggests that AZs in different compartments have distinct structures. Since individual KCs of each subtype synapse onto all compartments ^{8-10,40}, this Brp compartmental heterogeneity is likely formed intracellularly.

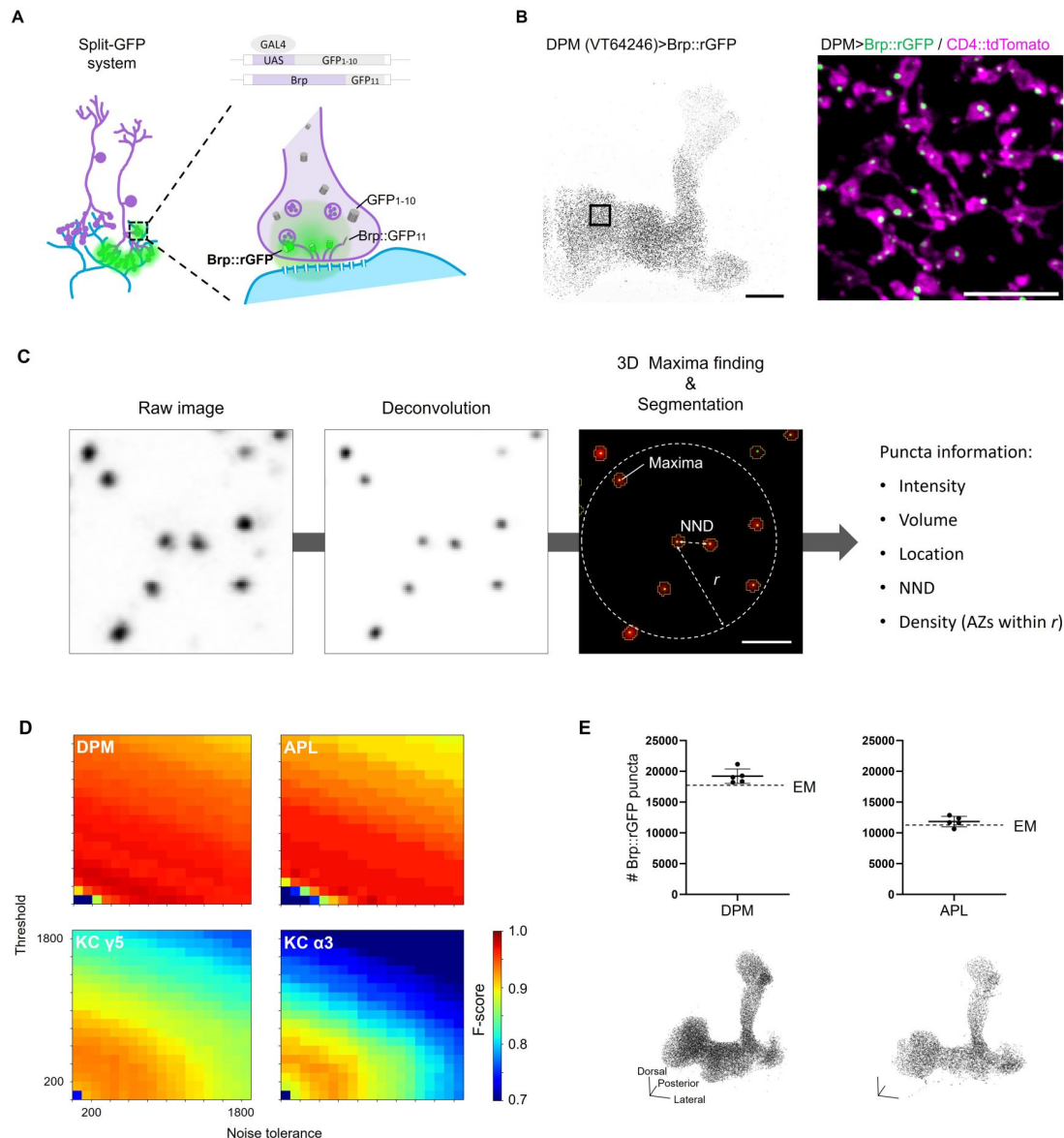


Fig. 1.

Spatial characterization of Brp::rGFP clusters at single-AZ level.

(A) Schematic of the split-GFP tagging strategy. The GFP₁₁ fragment is inserted by CRISPR/Cas9 just prior to the stop codon of *brp*, while GFP1-10 is expressed in specific cell types via GAL4-UAS system.

(B) Brp::rGFP signals in DPM. The entire MB lobe structure is shown. The black box in the left panel indicates the zoom-in area (right panel). GFP1-10 and CD4::tdTomato (magenta) were co-expressed using *VT64246-GAL4*. Scale bar, 20 μ m (left); 5 μ m (right).

(C) Image processing pipeline. Raw images were processed with image deconvolution to improve the signal quality. 3D maxima (green pixels) are detected for each Brp::rGFP cluster. Pixels around the maxima were clustered to create 3D ROIs, indicated by yellow circles. The "AZ density" is the number of Brp::rGFP clusters within r ($3 \times$ mean NND). Scale bar, 1 μ m.

(D) Heatmaps of F-scores showing the detection performance across in different cell types. An intensity threshold (threshold) is applied to reduce the background noise when detecting 3D maxima. The noise tolerance is the parameter that adjusts the sensitivity of 3D maxima detection. The color represents the F-score value.

(E) Number of Brp::rGFP clusters detected in DPM and APL. The dashed lines indicate the number of electron dense projections identified in the hemibrain connectome for DPM (17530) and APL (10789) on the MB lobes and peduncle (cite the resource here). For both DPM and APL, $n = 5$. Lower panels show the representative 3D reconstructions of Brp::rGFP clusters in DPM and APL. Scale bars, 20 μ m. Data were presented as mean $\pm$ s.e.m.

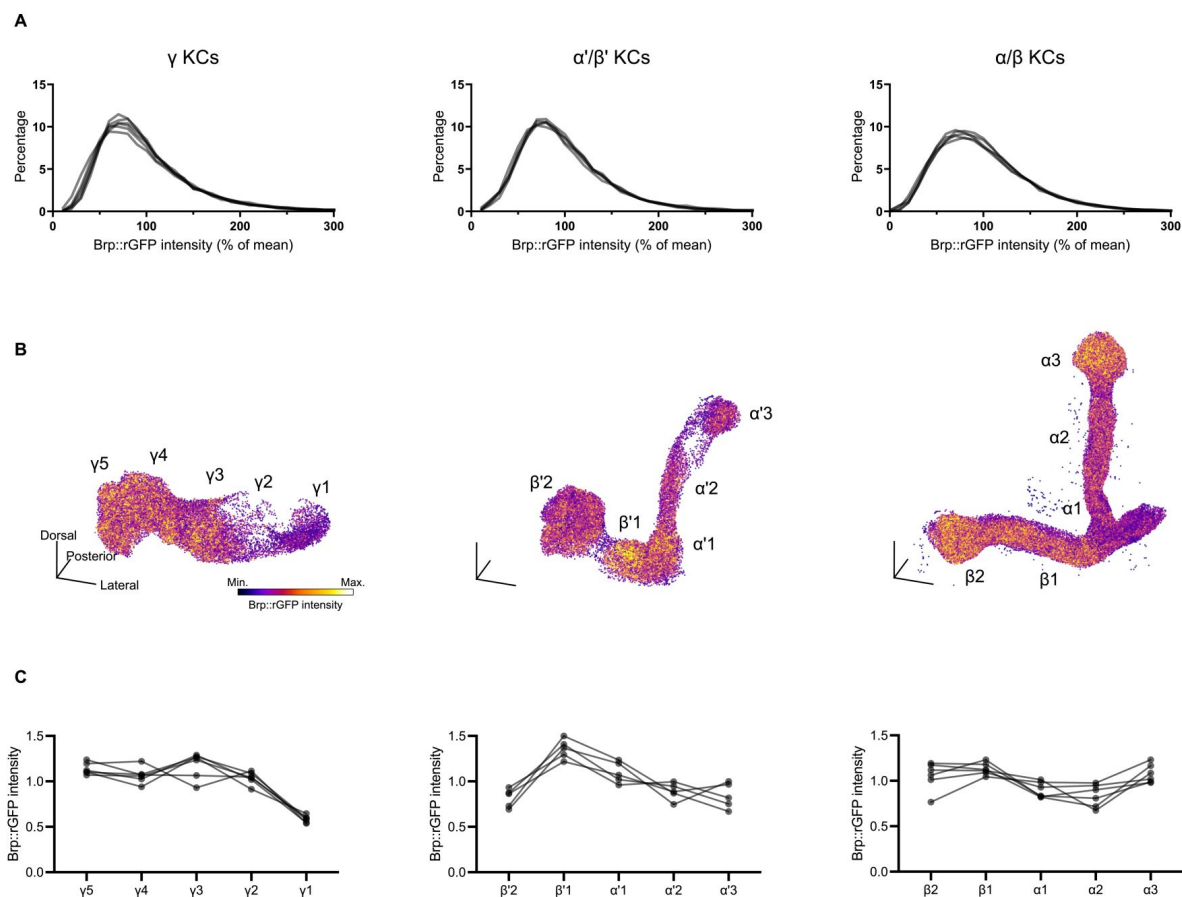


Fig. 2.

Compartmentalized AZ structures of KCs.

(A) Histograms of the signal intensity of individual Brp::rGFP clusters from three KC subtypes. Values are normalized by the mean intensity in each data set (each sample). Each line represents the histogram of one independent sample. γ KCs (*MB009B-GAL4*), $n = 6$; α/β KCs (*MB370B-GAL4*), $n = 5$; α'/β' KCs (*MB008B-GAL4*), $n = 5$.

(B) 3D reconstructions of Brp::rGFP clusters in three KC subtypes, colored by the Brp::rGFP intensity. The approximate locations of compartments are as indicated. Min. and Max. were set to represent the lowest and highest 5% of Brp::rGFP intensity value in the dataset respectively. Scale bars, 20 μ m.

(C) Signal intensity of Brp::rGFP clusters in each compartment. Brp::rGFP puncta were quantified compartmentally. Medians in different compartments are showed as the ratio against the average of five compartments in the corresponding KC subtype. Each line represents one independent sample.

Intracellular heterogeneity of Brp concentration at individual AZs and cell-type dependent diversity

Super-resolution microscopy has shown that the Brp molecular density at individual AZs is dynamically adjusted in motor neurons^{41,42}. We hypothesized that the amount of Brp does not necessarily correlate with AZ cluster size. To verify this, we calculated the Brp concentration by quantifying both the volume (number of voxels within a 3D ROI) and the Brp::rGFP intensity of each cluster. We found that clusters with similar sizes can exhibit vastly different signal intensities in the DPM neuron, suggesting the heterogeneity of Brp concentration at individual AZs (**Fig. 3A**). Correlation analysis between the Brp::rGFP intensity and cluster volume revealed a substantial variability among cell types (**Fig. 3B-E**). There were even significant differences among KC subtypes. These results suggest cell-type dependent variability of the Brp concentration of individual clusters.

Stereotypy of intracellular distribution of AZs

To study the spatial distribution of Brp clusters within single neurons, we focused on DANs. *MB504B-GAL4* labels four types of single PPL1 DANs projecting to distinct compartments⁴³. Using this driver, we analyzed Brp::rGFP distribution in each compartment and found it uneven in terminals of both PPL1- $\alpha 2 \alpha' 2$ and $\alpha 3$ (**Fig. 4A-B**). To better characterize cluster distribution, we calculated the “AZ density” defined as the number of Brp::rGFP clusters surrounding a particular cluster within a specified radius (see **Fig. 1A** and Materials and Methods for detail). By color-coding Brp::rGFP clusters according to their AZ density, we found that AZs in PPL1- $\alpha 3$ were more localized to the core of the $\alpha 3$ compartment. This pattern was consistent across individuals (**Fig. 4C-D**), suggesting a stereotyped AZ distribution within PPL1- $\alpha 3$ neurons.

We extended the analysis to two other single pairs of neurons innervating the MBs, the DPM and APL neurons. Visualizing AZ density revealed a stereotyped and compartment specific distribution in the DPM neuron (**Fig. 5A**). Specifically, the AZ density was constantly high in the α' / β' lobes across all individuals, supporting the known functional importance of DPM branches in the α' / β' lobes³⁰. In contrast, this compartmental pattern was less pronounced in the APL neuron. Notably, the ratio of AZ density between $\alpha 3$ and $\alpha' 3$ is strikingly different among individuals (**Fig. 5B**). Taken together, these data suggest that the individual variability in AZ distribution depends on the cell type.

Regulation of spatial configuration of AZs

Characteristics of Brp clusters and AZ distribution in KCs revealed spatial organization at the level of compartments. Recent studies further showed intra-compartmental variability of synaptic plasticity upon associative learning, prompting us to analyze sub-compartmental organization of AZs^{44,45}. We thus examined potential local patterns among AZs in proximity by calculating the correlation between nearest neighbor clusters in terms of their Brp::rGFP intensities. This analysis revealed strong correlations in KCs and APL but not in DPM (**Fig. 6A**).

We further analyzed the sub-compartmental AZ patterns by examining the spatial distribution of Brp::rGFP intensity. To reduce the high frequency variability of Brp::rGFP intensities (**Fig. 2B**), we calculated the “local intensity” for each cluster by applying mean filtering (the average Brp::rGFP intensity of all surrounding clusters within a specified radius r , see Materials and Methods for detail). By visualizing the local intensity in γ KCs, we identified AZ “hot spots”, sub-compartment-sized small groups of AZs with high local intensities, on top of the compartmental differences (**Fig. 6B**). Since Brp molecules are clustered more in AZ-dense boutons of a single motor neuron⁵, we hypothesized that these hot spots correspond to regions with higher AZ densities. Indeed, we found that hot spots appear in high AZ densities regions in γ KCs (**Fig. 6B**).

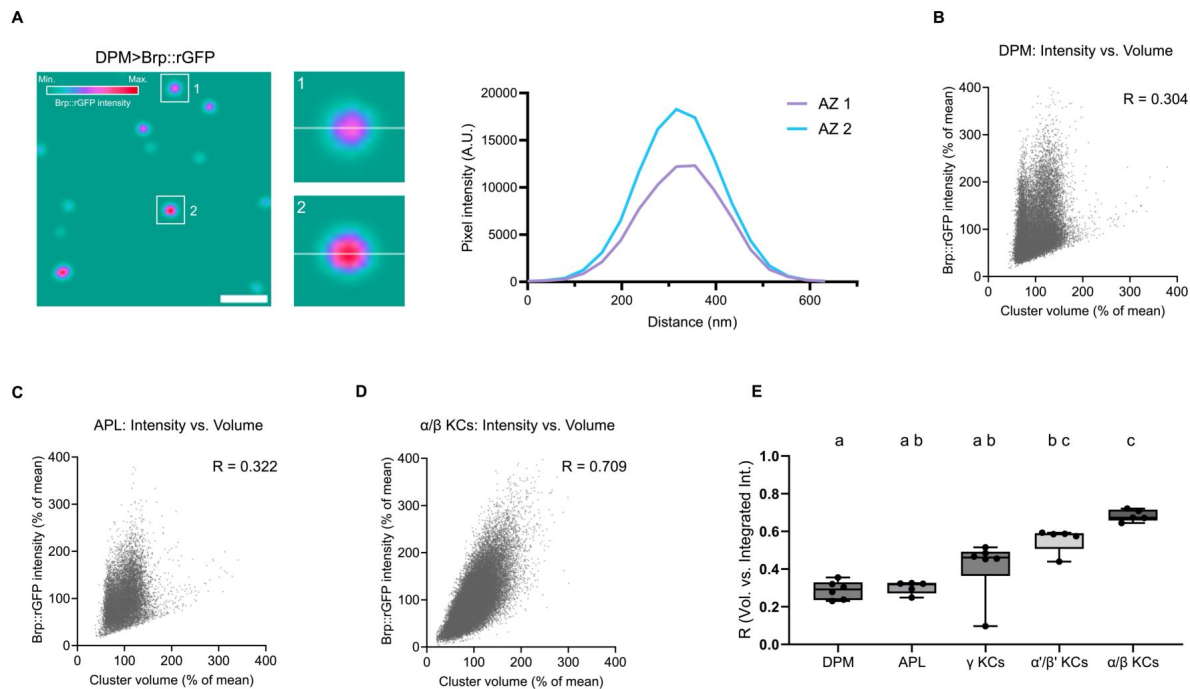


Fig. 3.

The variability of Brp::rGFP concentrations depends on cell type.

(A) Brp::rGFP clusters with distinct intensities but similar size in DPM. The left panel shows a cropped image of Brp::rGFP in DPM. White boxes indicate areas zoomed-in in panels 1 and 2. Transparent white lines in panels 1 and 2 show the lines on which the intensity profiles were plotted. Intensity profiles were plotted for AZ1 (magenta) and AZ2 (blue) respectively.

(B-D) Scatter plots showing the correlation between the Brp::rGFP intensity and cluster volume in DPM, APL and α/β KCs. Data of three representative samples were shown. Pearson correlation coefficient R were calculated for each sample.

(E) Correlations between the intensity and volume of Brp::rGFP clusters in different cell types. DPM ($n = 6$), APL ($n = 5$), γ KCs ($n = 6$, *MB009B-GAL4*), α'/β' KCs ($n = 5$, *MB370B-GAL4*), α/β KCs ($n = 5$, *MB008B-GAL4*). DPM vs. α'/β' KCs: $P = 0.0253$; DPM vs. α/β KCs: $P = 0.0009$; APL vs. α/β KCs: $P = 0.0035$; γ KCs vs. α/β KCs: $P = 0.0418$. Values marked with different lowercase letters represent significant difference ($P < 0.05$); Data were presented as box plots showing center (median), whiskers (Min. to Max.).

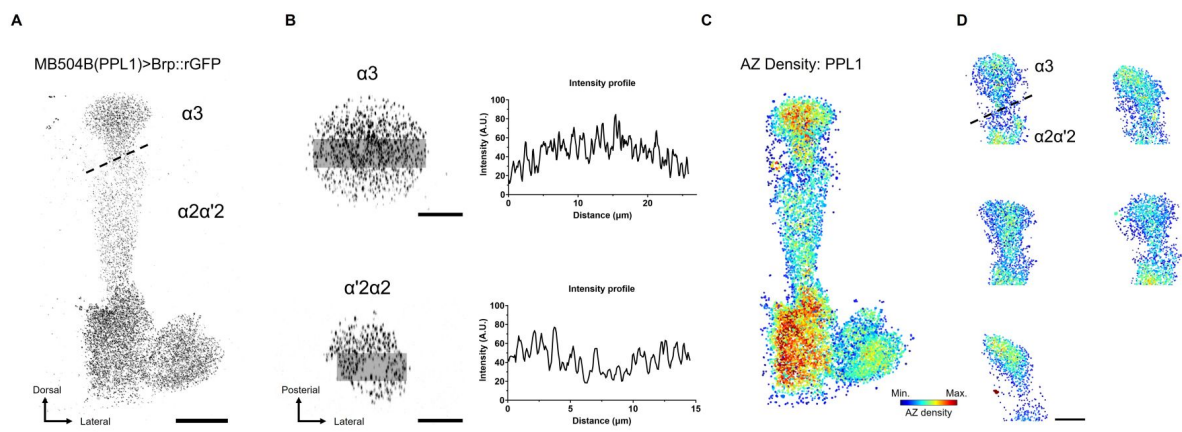


Fig. 4.

Stereotyped AZ distribution of PPL1-α3 DAN.

(A) Brp::rGFP in PPL1 DANs. UAS-GFP₁₋₁₀ was expressed using *MB504B-GAL4*. Dashed line marks the rough boundary between α3 and α2α'2 compartments. Scale bar, 20 μm.

(B) Brp::rGFP intensity profiles of α3 and α2α'2 compartments. Left panels show the max-projection images of α3 and α2α'2 optical coronal sections. Transparent gray stripes indicate areas where intensity profiles are plotted in the right panels. Scale bars, 10 μm.

(C) 3D reconstruction colored by the AZ density in PPL1 DANs. Color scale: Min. = 0, Max. = 40. Dashed line indicates the rough boundary between α3 and α2α'2 compartments. Scale bar, 20 μm.

(D) Stereotyped AZ distribution pattern in PPL1-α3 across individuals. 3D reconstructions show the AZ density across different brain samples. Scale bar, 20 μm.

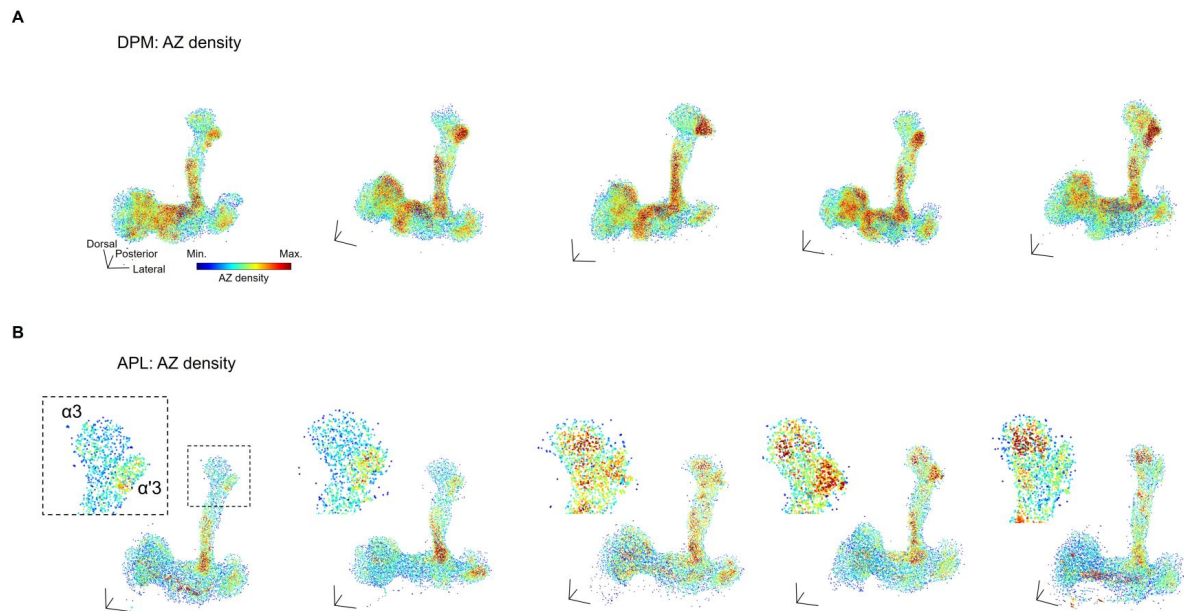


Fig. 5.

Cell-type-specific stereotypy of AZ spatial distributions.

(A) 3D reconstructions of Brp::rGFP clusters in DPMs, colored by AZ density. Color scale: Min. = 0, Max. = 35. Black arrows indicate consistently high AZ density regions across brain samples. Scale bars, 20 μ m.

(B) 3D reconstructions of Brp::rGFP clusters in APLs, colored by AZ density. Color scale: Min. = 0, Max. = 30. Black dashed square indicates the area zoomed-in. APL reconstructions are arranged from left to right according to the overall AZ density in $\alpha 3$.

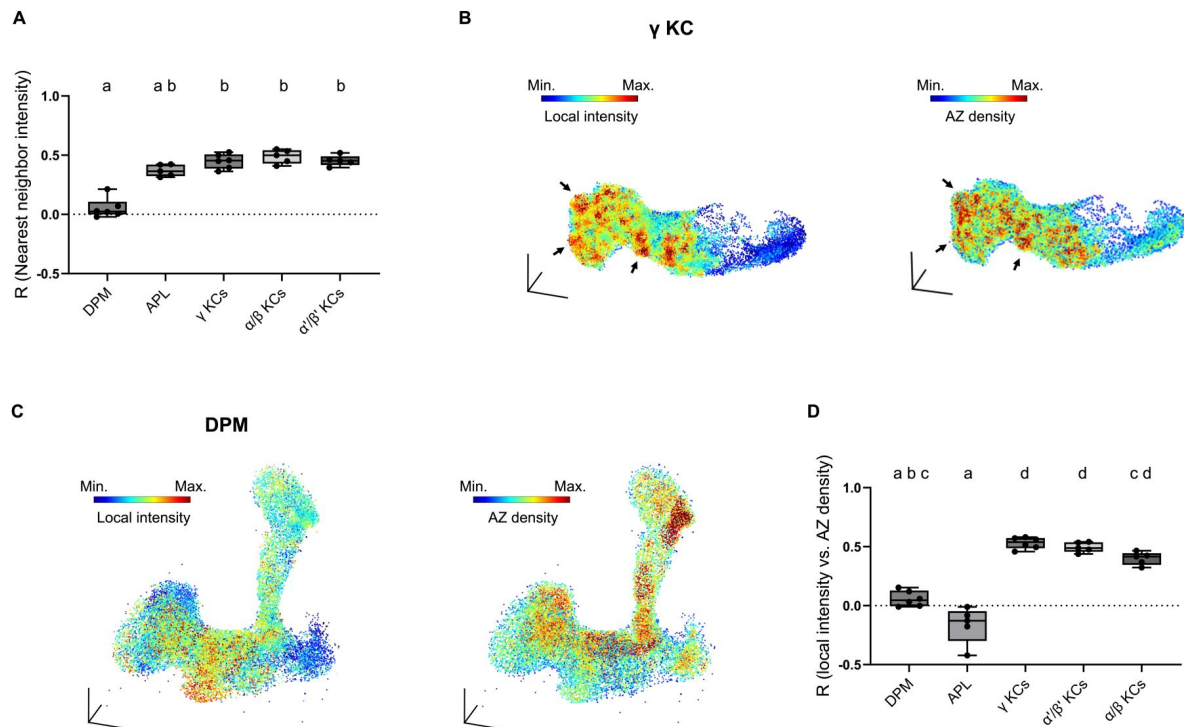


Fig. 6.

Local intensity analysis revealed sub-compartmental AZ structures.

(A) Correlation of Brp::rGFP intensities between nearest neighbor AZs in different cell types. Brp::rGFP intensities were log transformed. DPM vs. γ KCs: $P = 0.0066$; DPM vs. α'/β' KCs: $P = 0.0027$; DPM vs. α/β KCs: $P = 0.0066$; Scale bar: $1 \mu\text{m}$.

(B) 3D reconstructions of Brp::rGFP clusters in a γ KC sample, colored by local intensity and AZ density. Black arrows indicate areas with both high local intensities and high AZ densities. Min. and Max. were set to represent the lowest and highest 5% in local intensity value respectively. Color scale of AZ density: Min. = 0 and Max. = 35.

(C) 3D reconstructions of Brp::rGFP clusters in a DPM neuron, colored by local intensity and AZ density. Min. and Max. are set to represent the lowest and highest 5% in local intensity value respectively. Color scale of AZ density: Min. = 0, Max. = 35.

(D) Correlations between AZ density and local intensity in different cell types. DPM ($n = 6$), APL ($n = 5$), γ KCs ($n = 6$, *MB009B-GAL4*), α'/β' KCs ($n = 5$, *MB370B-GAL4*), α/β KCs ($n = 5$, *MB008B-GAL4*). DPM vs. γ KCs: $P = 0.0171$; DPM vs. α'/β' KCs: $P = 0.0171$; APL vs. γ KCs: $P = 0.0014$; APL vs. α'/β' KCs: $P = 0.0014$; APL vs. α/β KCs: $P = 0.0171$.

Consistently, correlation between the local intensity and the AZ density was high in all KC subtypes. In contrast, for DPM and APL neurons, the local intensity of Brp::rGFP was not associated with AZ density (**Fig. 6C-D**). Taken together, these results suggest previously unidentified sub-compartmental synaptic configuration organized across individual KCs.

Associative learning re-organizes sub-compartmental synaptic configuration

Sub-compartmental synaptic configuration may undergo experience-dependent changes, such as through associative learning^{17,18}. To examine such structural plasticity, we visualized Brp::rGFP specifically in KCs using *R13F02-GAL4* (**Fig. 7A**) and presented flies with odor (4-methylcyclohexanol) with a concomitant (paired) or shifted (unpaired) electric shock (**Fig. 7B**). The difference between the paired and unpaired groups therefore represents the effect of associative learning. We prepared the brains at different time points (ranging from 3 min to 1 day) after conditioning and measured the correlation between the AZ density and local intensity of individual Brp::rGFP clusters. Strikingly, we found that associative learning induced a transient modification to the KC synaptic configurations in a compartment-specific manner (**Fig. 7C**). This learning-induced structural plasticity was specific at 90 min after conditioning and disappeared within one day (**Fig. 7D**), consistent with previous studies reporting short-lived AZ remodeling in the MBs by conditioning^{17,18}. These results showcase the advantage of split-GFP system in reporting the dynamics of local synapse subsets with high throughputs and contents. Altogether, we suggest that associative learning induces transient memory traces of local structural plasticity in AZ configuration.

Discussion

By leveraging the CRISPR/Cas9 genome editing and split-GFP technique, we systematically profiled presynaptic structures of MB neurons in a cell-type-specific manner. This approach enabled high-throughput analysis of multiple brain samples and detailed characterizations, revealing previously inaccessible structural features of AZs in CNS neurons.

Single-cell level analysis of AZ profiles revealed cell-type-specific stereotypy in spatial regulation (**Fig. 4-6**). We found that AZ density in the DPM neurons is consistently high in the α'/β' lobes across individuals (**Fig. 5**). This AZ stereotypy of DPM explains the functional significance of the branches in the α'/β' lobes. Overexpression of a Down syndrome cell adhesion molecule (DScam) variant in the DPM neurons was shown to disrupt their innervation except in the α'/β' lobe, while memory formation remained unaffected³⁰. Similarly, our analysis identified the intracellular stereotypy in AZ distribution of the PPL1- $\alpha 3$ DAN within the $\alpha 3$ compartment (**Fig. 4**). This sub-compartmental organization of the single DAN and layered projections of α/β KCs may underlie the differential dopaminergic modulation of KC subtypes in associative learning⁴⁶. While such stereotypy is less pronounced in APL, we found strong correlations between Brp::rGFP intensities of neighboring AZs in contrast to DPM (**Fig. 6D**). This suggests that AZs in APL neurons are organized more locally, allowing them to modulate microcircuits composed of KCs, DANs and MBONs⁴⁷. Collectively, the spatial scale of AZ organizations depends on cell types, reflecting the function of individual MB neurons.

We observed that Brp::rGFP intensity correlates strongly with the AZ size in KC subtypes, in contrast to DPM and APL neurons (**Fig. 3**). Previous studies using localization microscopy showed that the molecular density of synaptic proteins at AZs can be dynamically adjusted in motor neurons, potentially modulating the synaptic vesicle release^{41,42,48}. The variable Brp concentrations (i.e. the ratio of the Brp::rGFP intensity to the AZ size) in DPM and APL neurons may represent AZs with distinct Brp densities and therefore different functional modes (**Fig. 3**)⁴¹. Notably, a recent study successfully predicted the neurotransmitter type of neurons using the

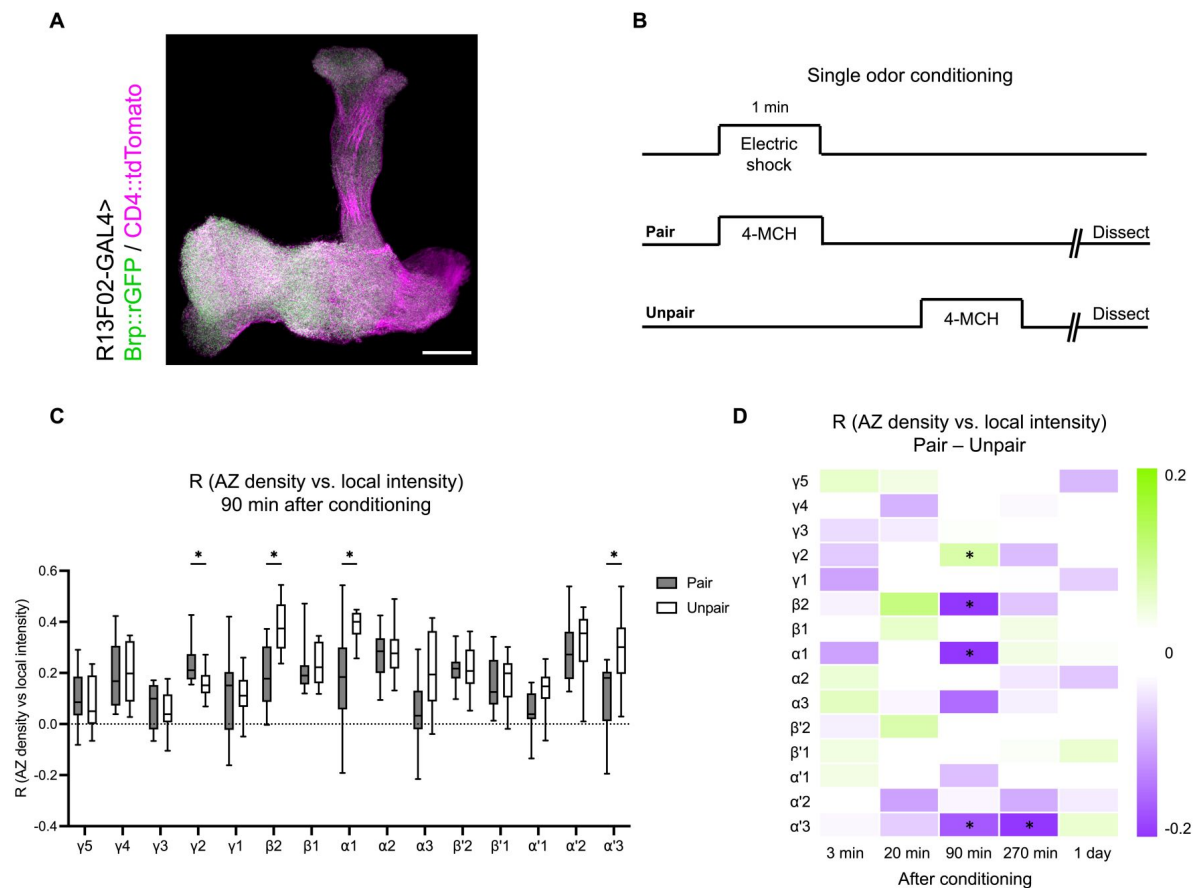


Fig. 7.

Associative conditioning re-organizes sub-compartmental active zone clusters.

(A) Brp::rGFP (green) and CD4::tdTomato cytoplasm membrane marker (magenta) in KCs, visualized by using *R13F02-GAL4*.

(B) The experimental design of the aversive single odor conditioning. Flies in the pair group receive concurrent presentation of 2% 4-MCH and 90 V electric shock. Flies in the unpair group first receive the electric shock first and then 4-MCH 1 min later. Flies were dissected at different time points after the conditioning.

(C) Correlation coefficient (AZ density vs. local intensity of individual Brp::rGFP puncta) of each compartment at 90 min after conditioning. Pair (n = 11) vs. Unpair (n = 12), y2 (P = 0.0261), β 2 (P = 0.0060), α 1 (P = 0.0026), α '3 (P = 0.0026).

(D) Heatmap showing the difference of correlation coefficient (AZ density vs. local intensity of Brp::rGFP puncta) between the pair and unpair group at different time points after conditioning. The color indicates the difference and the asterisks in the compartments indicate the significant difference. For 3 min, Pair (n = 12), Unpair (n = 11); For 20 min, Pair (n = 12), Unpair (n = 11); For 90 min, Pair (n = 11) vs. Unpair (n = 12), P = 0.0247; For 270 min, Pair (n = 11), Unpair (n = 11); For 1 D, Pair (n = 12), Unpair (n = 9); For all results involved statistical comparison, only significant results are shown. Scale bars, 20 μ m. Mann-Whitney test with original False Discovery Rate (FDR) method of Benjamini and Hochberg correction. *P<0.05. Data were presented as box plots showing centre (median), whiskers (Min. to Max.).

ultrastructure of dense projections as one of the parameters⁴⁹, suggesting neurotransmitter-specific AZ structures. Interestingly, both the DPM and APL neurons are reported to release multiple neurotransmitters^{27,31,50–53}. Distinct Brp concentrations at AZs might thus support particular release modes or neurotransmitter types.

In KCs, the spatial analysis of local AZ configuration revealed strong correlations between Brp::rGFP intensities at neighboring AZs as well as AZ “hot spots”, regions with high Brp localization and AZ densities (**Fig. 6**). These results suggest previously unidentified, highly localized sub-compartmental AZ structures, which are likely organized across individual KCs. Presynaptic terminals of a single KC are found to be highly heterogeneous in terms of their plasticity upon dopamine modulation and associative learning^{44,45}. A recent study found the sub-compartmentally heterogeneous activities of dopamine terminals explain memory specificity²⁴. Such locally distinct dopamine input may be a source of the AZ hot spot formation in KC terminals. This model is consistent with the localization of two opposing dopamine receptors Dop1R1 and Dop2R to AZs⁵⁴. Furthermore, we showed that the local configuration undergoes structural plasticity upon associative learning that involves presynaptic dopaminergic modulation^{55–60} (**Fig. 7**). Since cAMP signaling plays a crucial role in the AZ structural plasticity especially during associative memory formation^{39,48,61}, it may underlie learning-induced plasticity in sub-compartmental AZ configuration^{17,18}.

Materials and Methods

Animals

Flies were maintained on standard cornmeal food at 25 °C under a 12:12 h light-dark cycle for all experiments. All flies used for experiments are 3-7 days old adult males. Flies were transferred to fresh food vials after hatching and flipped every two days before experiments. The GAL4-UAS system was used to express transgenes, and balancers were removed for animals used in all experiments. Fly strains and resources used in this study are as follow: *MB008B-GAL4* (BDSC 68291)²³, *MB009B-GAL4* (BDSC 68292)²³, *MB370B-GAL4* (BDSC 68319)²³, *MB504B-GAL4* (BDSC 68329)²³, *VT64246-GAL4* (VDRC 204311)⁶², *GH146-GAL4* (BDSC 30026)⁶³, *R58E02-GAL4* (BDSC 41347)⁶⁴, *R13F02-GAL4* (BDSC 48571)⁶⁴, *UAS-CD4::tdTomato* (BDSC 35841, by Dr. Yhu Nung Jan & Chun Han),⁵⁹ *brp::GFP₁₁*³⁹, *UAS-GFP₁₋₁₀*³⁵.

Single-odor aversive conditioning

Adult flies at 3-7 days after eclosion were kept in a 25 °C incubator before and after conditioning. Approximately 50 flies were put in a training tube that is set inside the climate box for 2 minutes before the conditioning. Temperature in the climate box was set at 24 °C and the relative humidity was set to 70%. For the paired group, 2% 4-methylcyclohexanol (4-MCH, Sigma-Aldrich, diluted in paraffin oil, Sigma-Aldrich) was presented in a cup with the diameter of 5 mm, together with 12 pulses of electric shocks at 90 V for 1 minute inside the training tube. The unpaired group received the same 1-minute electric shock first, followed by 1-minute rest, and then the same 1-minute 4-MCH presentation. Flies were dissected at 3 min, 20 min, 90 min, 270 min and 1 day after conditioning. Experiments were performed multiple times on different days for a single group.

Sample preparation

Animals used in experiments involving comparisons between control and experimental groups were dissected on the same day. Data were collected from multiple batches of experiments performed on different days. Flies were anesthetized by ice and placed on ice before dissection. The dissection was performed in cold PBS to collect the brain without the ventral nerve cord. The

brains were subsequently fixed by 2% paraformaldehyde for 1 h and then washed 3×20 min by 0.1% PBT (0.1% Triton X-100 in PBS) in a PCR tube, typically 5-6 brains in one tube. Samples were mounted on microscope slides using mounting medium SeeDB2S ⁶⁵ [\[link\]](#).

Confocal microscopy image acquisition

The image acquisition was performed using an Olympus FV1200 confocal microscope platform equipped with both PMT and GaAsP high sensitivity detectors and a 60×/1.42 NA oil immersion objective (PLAPON60XO, Olympus). The Kalman filter was turned on to 2× averaging while scanning. For Brp::rGFP samples in all GAL4 types, scanning speed was set at 2 μs/pixel with a voxel size of lateral 0.079 × 0.079 × axial 0.370 μm. For experimental point spread function (PSF) imaging, SeeDB2S immersed beads were scanned in a setting that is 60×/1.42 NA oil immersion objective; 473 nm laser power: 2.0%; 559 nm laser power: 1.5%; voxel size: 0.079 μm × 0.079 μm × 0.370 μm; scanning speed: 4 μs/pixel, to produce multiple bead images.

PSF acquisition and image deconvolution

Sub-resolution fluorescent beads (Tetraspeck Microspheres 0.1 μm, Thermo Fisher Scientific, T7279) were imaged to generate the PSF. The bead solution was diluted with distilled water and sonicated several times to eliminate aggregation. PSF images acquired were processed using Amira software (Thermo Fisher Scientific) with the Extract Point Spread Function module. The PSFs extracted from each image were averaged to create a single PSF for later image deconvolution (Resizing voxel size: 0.079 μm × 0.079 μm × 0.370 μm; 32 pixels × 32 pixels × 21 slices of image size). Image deconvolution was performed by using the Richardson-Lucy iterative non-blind algorithm in a Fiji plugin DeconvolutionLab2 ⁶⁶ [\[link\]](#), or by CLIJ2 GPU-based Richardson-Lucy deconvolution ⁶⁷ [\[link\]](#). Images were deconvolved 20 times iteratively to improve image quality.

Brp clusters segmentation and data analysis

For Brp::rGFP cluster detection, the image stack size was adjusted to double the XY plate pixel number using Fiji to increase the detection precision. The images were subsequently processed using the 3D maxima finder and the 3D spot segmentation function of the 3D suite package in Fiji. 3D maxima were identified for each cluster and used as the starting point for pixel clustering ³⁶ [\[link\]](#). An intensity threshold (threshold) and the noise tolerance were set to reduce noise and adjust the sensitivity of 3D maxima detection respectively. The threshold set the value of all pixels that have values below the threshold to NaN (not a number). The noise tolerance is set in the 3D maxima finder. To be noticed, the threshold is only applied to the image used to find 3D maxima. The 3D maxima finder generates the 3D maxima. The 3D segmentation function then requires the 3D maxima and the raw image, so that there is no intensity cut affecting the measurement of cluster volume (used deconvolved image). The local thresholding method for the 3D segmentation was set to Gaussian fit, with the “SD value” set to 2. Cluster volume was measured as the voxel count within the 3D ROI.

The above procedures may include a tiling and stitching step for images and data because of the limited computer power. Typically, an image was tiled into 25 sub-stacks with overlapped margins. After being processed separately, the data acquired from all tiled sub-stacks was stitched back with a deletion of overlapping data points to generate the complete detection result. Nearest neighbors were identified for each particle and spatial distances in between (NND) were calculated using Python. For each sample, mean NND was calculated separately. For a particular Brp::rGFP cluster, the number of surrounding clusters within r ($r = 3 \times \text{mean NND}$) spherically including itself is referred as the “AZ density”. The “local intensity” of a particular Brp::rGFP cluster is the average of Brp::rGFP intensity of all clusters surrounding within r ($r = 3 \times \text{mean NND}$) including itself. The plot function of Python was used to plot the color-coded 3D reconstructed maps.

F-score calculation

The 3D detection parameters were optimized by matching the 3D maxima detection results to the ground truths. Ground truths were defined by annotating AZs manually, according to the size, position and signal pattern of Brp puncta in the images. The F-score was used to assess detection precision. F-score is the harmonic mean of a system's precision and recall value, calculated by the following formula: $2 \times [(Precision \times Recall) / (Precision + Recall)]$. The precision is the number of true positives divided by the total number of identified positives, including incorrect identification. The recall is the number of true positives divided by the total number of true data points defined as the ground truth. F-scores shown in **Fig. 1** were automatically calculated using a customized Fiji Macro that aligned 3D maxima with ground truth. Different combinations of noise threshold and noise tolerance were tested for the test images cropped from raw images.

Statistical analysis

Statistical analysis was performed using GraphPad Prism version 8, 9 and 10 or Python. For statistical comparisons, Kruskal-Wallis test with original False Discovery Rate method of Benjamini and Hochberg correction was applied except as explicitly mentioned. A desired false discovery rate of 0.05. Pearson's correlation coefficient (R) was calculated using Python.

Data and materials availability

Further information and requests on resources and reagents should be directed to and will be fulfilled by the corresponding author, Dr. Hiromu Tanimoto (hiromut@m.tohoku.ac.jp). This study did not generate any unique code. All data needed to evaluate the conclusions in the paper are present in the paper.

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Visualization: HW, SE

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Writing—original draft: HY, HT

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References

1. Kittel R.J., Wichmann C., Rasse T.M., Fouquet W., Schmidt M., Schmid A., Wagh D.A., Pawlu C., Kellner R.R., Willig K.I., et al. (2006) **Bruchpilot Promotes Active Zone Assembly, Ca²⁺ Channel Clustering, and Vesicle Release** *Science* **312**:1051–1054 <https://doi.org/10.1126/science.1126308> | Google Scholar
2. Wagh D.A., Rasse T.M., Asan E., Hofbauer A., Schwenkert I., Dürrebeck H., Buchner S., Dabauvalle M.-C., Schmidt M., Qin G., et al. (2006) **Bruchpilot, a Protein with Homology to ELKS/CAST, Is Required for Structural Integrity and Function of Synaptic Active Zones in Drosophila** *Neuron* **49**:833–844 <https://doi.org/10.1016/j.neuron.2006.02.008> | Google Scholar
3. Fouquet W., Oswald D., Wichmann C., Mertel S., Depner H., Dyba M., Hallermann S., Kittel R.J., Eimer S., Sigrist S.J (2009) **Maturation of active zone assembly by Drosophila Bruchpilot** *J Cell Biol* **186**:129–145 <https://doi.org/10.1083/jcb.200812150> | Google Scholar
4. Matkovic T., Siebert M., Knoche E., Depner H., Mertel S., Oswald D., Schmidt M., Thomas U., Sickmann A., Kamin D., et al. (2013) **The Bruchpilot cytomatrix determines the size of the readily releasable pool of synaptic vesicles** *J Cell Biol* **202**:667–683 <https://doi.org/10.1083/jcb.201301072> | Google Scholar
5. Paul M.M., Pauli M., Ehmann N., Hallermann S., Sauer M., Kittel R.J., Heckmann M (2015) **Bruchpilot and Synaptotagmin collaborate to drive rapid glutamate release and active zone differentiation** *Front Cell Neurosci* **9**:29 <https://doi.org/10.3389/fncel.2015.00029> | Google Scholar
6. Akbergenova Y., Cunningham K.L., Zhang Y.V., Weiss S., Littleton J.T (2018) **Characterization of developmental and molecular factors underlying release heterogeneity at Drosophila synapses** *eLife* **7**:e38268 <https://doi.org/10.7554/elife.38268> | Google Scholar
7. Newman Z.L., Bakshinskaya D., Schultz R., Kenny S.J., Moon S., Aghi K., Stanley C., Marnani N., Li R., Bleier J., et al. (2022) **Determinants of synapse diversity revealed by super-resolution quantal transmission and active zone imaging** *Nat Commun* **13**:229 <https://doi.org/10.1038/s41467-021-27815-2> | Google Scholar
8. Zheng Z., Lauritzen J.S., Perlman E., Robinson C.G., Nichols M., Milkie D., Torrens O., Price J., Fisher C.B., Sharifi N., et al. (2018) **A Complete Electron Microscopy Volume of the Brain of Adult Drosophila melanogaster** *Cell* **174**:730–743 <https://doi.org/10.1016/j.cell.2018.06.019> | Google Scholar
9. Scheffer L.K., Xu C.S., Januszewski M., Lu Z., Takemura S., Hayworth K.J., Huang G.B., Shinomiya K., Maitlin-Shepard J., Berg S., et al. (2020) **A connectome and analysis of the adult Drosophila central brain** *eLife* **9**:e57443 <https://doi.org/10.7554/elife.57443> | Google Scholar
10. Takemura S., Hayworth K.J., Huang G.B., Januszewski M., Lu Z., Marin E.C., Preibisch S., Xu C.S., Bogovic J., Champion A.S., et al. (2024) **A Connectome of the Male Drosophila Ventral Nerve Cord** *eLife* **13**:RP97769 <https://doi.org/10.7554/elife.97769.1> | Google Scholar
11. Schlegel P., Yin Y., Bates A.S., Dorkenwald S., Eichler K., Brooks P., Han D.S., Gkantia M., Santos M. dos, Munnelly E.J., et al. (2024) **Whole-brain annotation and multi-connectome cell**

typing of *Drosophila* *Nature* **634**:139–152 <https://doi.org/10.1038/s41586-024-07686-5> | [Google Scholar](#)

12. Dorkenwald S., Matsliah A., Sterling A.R., Schlegel P., Yu S., McKellar C.E., Lin A., Costa M., Eichler K., Yin Y., et al. (2024) **Neuronal wiring diagram of an adult brain** *Nature* **634**:124–138 <https://doi.org/10.1038/s41586-024-07558-y> | [Google Scholar](#)
13. Sachse S., Rueckert E., Keller A., Okada R., Tanaka N.K., Ito K., Vosshall L.B (2007) **Activity-Dependent Plasticity in an Olfactory Circuit** *Neuron* **56**:838–850 <https://doi.org/10.1016/j.neuron.2007.10.035> | [Google Scholar](#)
14. Gilestro G.F., Tononi G., Cirelli C (2009) **Widespread Changes in Synaptic Markers as a Function of Sleep and Wakefulness in *Drosophila*** *Science* **324**:109–112 <https://doi.org/10.1126/science.1166673> | [Google Scholar](#)
15. Fernández M.P., Berni J., Ceriani M.F (2008) **Circadian Remodeling of Neuronal Circuits Involved in Rhythmic Behavior** *PLoS Biol* **6**:e69 <https://doi.org/10.1371/journal.pbio.0060069> | [Google Scholar](#)
16. Kremer M.C., Christiansen F., Leiss F., Paehler M., Knappek S., Andlauer T.F.M., Förstner F., Kloppenburg P., Sigrist S.J., Tavosanis G (2010) **Structural Long-Term Changes at Mushroom Body Input Synapses** *Curr Biol* **20**:1938–1944 <https://doi.org/10.1016/j.cub.2010.09.060> | [Google Scholar](#)
17. Zhang X., Li Q., Wang L., Liu Z.-J., Zhong Y (2018) **Active Protection: Learning-Activated Raf/MAPK Activity Protects Labile Memory from Rac1-Independent Forgetting** *Neuron* **98**:142–155 <https://doi.org/10.1016/j.neuron.2018.02.025> | [Google Scholar](#)
18. Turrel O., Ramesh N., Escher M.J.F., Pooryasin A., Sigrist S.J (2022) **Transient active zone remodeling in the *Drosophila* mushroom body supports memory** *Curr Biol* <https://doi.org/10.1016/j.cub.2022.10.017> | [Google Scholar](#)
19. Mosca T.J., Luo L (2014) **Synaptic organization of the *Drosophila* antennal lobe and its regulation by the Teneurins** *eLife* **3**:e03726 <https://doi.org/10.7554/elife.03726> | [Google Scholar](#)
20. Gao R., Asano S.M., Upadhyayula S., Pisarev I., Milkie D.E., Liu T.-L., Singh V., Graves A., Huynh G.H., Zhao Y., et al. (2019) **Cortical column and whole-brain imaging with molecular contrast and nanoscale resolution** *Science* **363** <https://doi.org/10.1126/science.aau8302> | [Google Scholar](#)
21. Chen Y., Akin O., Nern A., Tsui C.Y.K., Pecot M.Y., Zipursky S.L (2014) **Cell-type-Specific Labeling of Synapses In Vivo through Synaptic Tagging with Recombination** *Neuron* **81**:280–293 <https://doi.org/10.1016/j.neuron.2013.12.021> | [Google Scholar](#)
22. Davis R.L (2023) **Learning and memory using *Drosophila melanogaster* : a focus on advances made in the fifth decade of research** *Genetics* <https://doi.org/10.1093/genetics/iyad085> | [Google Scholar](#)
23. Aso Y., Hattori D., Yu Y., Johnston R.M., Iyer N.A., Ngo T.-T., Dionne H., Abbott L., Axel R., Tanimoto H., et al. (2014) **The neuronal architecture of the mushroom body provides a**

logic for associative learning *eLife* **3**:e04577 <https://doi.org/10.7554/elife.04577> | Google Scholar

24. Yamagata N., Ezaki T., Takahashi T., Wu H., Tanimoto H (2021) **Presynaptic inhibition of dopamine neurons controls optimistic bias** *eLife* **10**:e64907 <https://doi.org/10.7554/elife.64907> | Google Scholar
25. Pitman J.L., Huetteroth W., Burke C.J., Krashes M.J., Lai S.-L., Lee T., Waddell S (2011) **A Pair of Inhibitory Neurons Are Required to Sustain Labile Memory in the Drosophila Mushroom Body** *Curr Biol* **21**:855–861 <https://doi.org/10.1016/j.cub.2011.03.069> | Google Scholar
26. Lin A.C., Bygrave A.M., Calignon A. de, Lee T., Miesenböck G. (2014) **Sparse, decorrelated odor coding in the mushroom body enhances learned odor discrimination** *Nat Neurosci* **17**:559–568 <https://doi.org/10.1038/nn.3660> | Google Scholar
27. Liu X., Davis R.L (2009) **The GABAergic anterior paired lateral neuron suppresses and is suppressed by olfactory learning** *Nat. Neurosci* **12**:53–59 <https://doi.org/10.1038/nn.2235> | Google Scholar
28. Yu D., Keene A.C., Srivatsan A., Waddell S., Davis R.L (2005) **Drosophila DPM Neurons Form a Delayed and Branch-Specific Memory Trace after Olfactory Classical Conditioning** *Cell* **123**:945–957 <https://doi.org/10.1016/j.cell.2005.09.037> | Google Scholar
29. Keene A.C., Stratmann M., Keller A., Perrat P.N., Vosshall L.B., Waddell S (2004) **Diverse Odor-Conditioned Memories Require Uniquely Timed Dorsal Paired Medial Neuron Output** *Neuron* **44**:521–533 <https://doi.org/10.1016/j.neuron.2004.10.006> | Google Scholar
30. Keene A.C., Krashes M.J., Leung B., Bernard J.A., Waddell S (2006) **Drosophila Dorsal Paired Medial Neurons Provide a General Mechanism for Memory Consolidation** *Curr Biol* **16**:1524–1530 <https://doi.org/10.1016/j.cub.2006.06.022> | Google Scholar
31. Haynes P.R., Christmann B.L., Griffith L.C (2015) **A single pair of neurons links sleep to memory consolidation in Drosophila melanogaster** *eLife* **4**:e03868 <https://doi.org/10.7554/elife.03868> | Google Scholar
32. Waddell S., Armstrong J.D., Kitamoto T., Kaiser K., Quinn W.G (2000) **The amnesiac Gene Product Is Expressed in Two Neurons in the Drosophila Brain that Are Critical for Memory** *Cell* **103**:805–813 [https://doi.org/10.1016/S0092-8674\(00\)00183-5](https://doi.org/10.1016/S0092-8674(00)00183-5) | Google Scholar
33. Huang S., Piao C., Beuschel C.B., Götz T., Sigrist S.J (2020) **Presynaptic Active Zone Plasticity Encodes Sleep Need in Drosophila** *Curr Biol* **30**:1077–1091 <https://doi.org/10.1016/j.cub.2020.01.019> | Google Scholar
34. Kamiyama D., Sekine S., Barsi-Rhyne B., Hu J., Chen B., Gilbert L.A., Ishikawa H., Leonetti M.D., Marshall W.F., Weissman J.S., et al. (2016) **Versatile protein tagging in cells with split fluorescent protein** *Nat. Commun* **7**:11046 <https://doi.org/10.1038/ncomms11046> | Google Scholar
35. Kondo S., Takahashi T., Yamagata N., Imanishi Y., Katow H., Hiramatsu S., Lynn K., Abe A., Kumaraswamy A., Tanimoto H (2020) **Neurochemical Organization of the Drosophila Brain Visualized by Endogenously Tagged Neurotransmitter Receptors** *Cell Reports* **30**:284–297 <https://doi.org/10.1016/j.celrep.2019.12.018> | Google Scholar

36. Ollion J., Cochennec J., Loll F., Escudé C., Boudier T (2013) **TANGO: a generic tool for high-throughput 3D image analysis for studying nuclear organization** *Bioinformatics* **29**:1840–1841 <https://doi.org/10.1093/bioinformatics/btt276> | Google Scholar
37. Ehmann N., Linde S. van de, Alon A., Ljaschenko D., Keung X.Z., Holm T., Rings A., DiAntonio A., Hallermann S., Ashery U., et al. (2014) **Quantitative super-resolution imaging of Bruchpilot distinguishes active zone states** *Nat Commun* **5**:4650 <https://doi.org/10.1038/ncomms5650> | Google Scholar
38. Gratz S.J., Goel P., Bruckner J.J., Hernandez R.X., Khateeb K., Macleod G.T., Dickman D., O'Connor-Giles K.M (2019) **Endogenous tagging reveals differential regulation of Ca²⁺ channels at single AZs during presynaptic homeostatic potentiation and depression** *J Neurosci* **39**:3068–18 <https://doi.org/10.1523/jneurosci.3068-18.2019> | Google Scholar
39. Wu H., Eno S., Jinnai K., Maekawa Y., Saito K., Abe A., Williams D.W., Yamagata N., Kondo S., Tanimoto H (2024) **Octopamine signals coordinate the spatial pattern of presynaptic machineries within individual Drosophila mushroom body neurons** *bioRxiv* :12 <https://doi.org/10.1101/2024.12.02.626512> | Google Scholar
40. Schlegel P., Yin Y., Bates A.S., Dorkenwald S., Eichler K., Brooks P., Han D.S., Gkantia M., et al. (2023) **A consensus cell type atlas from multiple connectomes reveals principles of circuit stereotypy and variation** *bioRxiv* <https://doi.org/10.1101/2023.06.27.546055> | Google Scholar
41. Mrestani A., Pauli M., Kollmannsberger P., Repp F., Kittel R.J., Eilers J., Doose S., Sauer M., Sirén A.-L., Heckmann M., et al. (2021) **Active zone compaction correlates with presynaptic homeostatic potentiation** *Cell Rep* **37**:109770 <https://doi.org/10.1016/j.celrep.2021.109770> | Google Scholar
42. Ghelani T., Escher M., Thomas U., Esch K., Lützkendorf J., Depner H., Maglione M., Parutto P., Gratz S., Matkovic-Rachid T., et al. (2023) **Interactive nanocluster compaction of the ELKS scaffold and Cacophony Ca²⁺ channels drives sustained active zone potentiation** *Sci. Adv* **9**:eade7804 <https://doi.org/10.1126/sciadv.ade7804> | Google Scholar
43. Vogt K., Schnaitmann C., Dylla K.V., Knappek S., Aso Y., Rubin G.M., Tanimoto H (2014) **Shared mushroom body circuits underlie visual and olfactory memories in Drosophila** *eLife* **3**:e02395 <https://doi.org/10.7554/elife.02395> | Google Scholar
44. Davidson A.M., Kaushik S., Hige T (2023) **Dopamine-Dependent Plasticity Is Heterogeneously Expressed by Presynaptic Calcium Activity across Individual Boutons of the Drosophila Mushroom Body** *eNeuro* **10**:ENEURO.0275-23.2023 <https://doi.org/10.1523/jneuro.0275-23.2023> | Google Scholar
45. Bilz F., Geurten B.R.H., Hancock C.E., Widmann A., Fiala A (2020) **Visualization of a Distributed Synaptic Memory Code in the Drosophila Brain** *Neuron* **106**:963–976 <https://doi.org/10.1016/j.neuron.2020.03.010> | Google Scholar
46. Perisse E., Yin Y., Lin A.C., Lin S., Huetteroth W., Waddell S (2013) **Different Kenyon Cell Populations Drive Learned Approach and Avoidance in Drosophila** *Neuron* **79**:945–956 <https://doi.org/10.1016/j.neuron.2013.07.045> | Google Scholar
47. Amin H., Apostolopoulou A.A., Suárez-Grimalt R., Vrontou E., Lin A.C (2020) **Localized inhibition in the Drosophila mushroom body** *eLife* **9**:e56954 <https://doi.org/10.7554/elife.56954> | Google Scholar

48. Sachidanandan D., Aravamudhan A., Mrestani A., Nerlich J., Lamberty M., Hasenauer N., Ehmann N., Pauls D., Seubert T., Maiellaro I., et al. (2023) **Rab3 mediates cyclic AMP-dependent presynaptic plasticity and olfactory learning** *bioRxiv* <https://doi.org/10.1101/2023.12.21.572589> | Google Scholar
49. Eckstein N., Bates A.S., Champion A., Du M., Yin Y., Schlegel P., Lu A.K.-Y., Rymer T., Finley-May S., Paterson T., et al. (2024) **Neurotransmitter classification from electron microscopy images at synaptic sites in *Drosophila melanogaster*** *Cell* **187**:2574–2594 <https://doi.org/10.1016/j.cell.2024.03.016> | Google Scholar
50. Wu C.-L., Shih M.-F.M., Lee P.-T., Chiang A.-S (2013) **An Octopamine-Mushroom Body Circuit Modulates the Formation of Anesthesia-Resistant Memory in *Drosophila*** *Curr Biol* **23**:2346–2354 <https://doi.org/10.1016/j.cub.2013.09.056> | Google Scholar
51. Lee P.-T., Lin H.-W., Chang Y.-H., Fu T.-F., Dubnau J., Hirsh J., Lee T., Chiang A.-S (2011) **Serotonin-mushroom body circuit modulating the formation of anesthesia-resistant memory in *Drosophila*** *Proc. Natl. Acad. Sci* **108**:13794–13799 <https://doi.org/10.1073/pnas.1019483108> | Google Scholar
52. Davie K., Janssens J., Koldere D., Waegeneer M.D., Pech U., Kreft Ł., Aibar S., Makhzami S., Christiaens V., González-Blas C.B., et al. (2018) **A Single-Cell Transcriptome Atlas of the Aging *Drosophila* Brain** *Cell* **174**:982–998 <https://doi.org/10.1016/j.cell.2018.05.057> | Google Scholar
53. Zeng J., Li X., Zhang R., Lv M., Wang Y., Tan K., Xia X., Wan J., Jing M., Zhang X., et al. (2023) **Local 5-HT signaling bi-directionally regulates the coincidence time window for associative learning** *Neuron* **111**:1118–1135 <https://doi.org/10.1016/j.neuron.2022.12.034> | Google Scholar
54. Hiramatsu S., Saito K., Kondo S., Katow H., Yamagata N., Wu C.-F., Tanimoto H (2024) **Cell-type-specific fluorescent tagging of endogenous target proteins reveals synaptic enrichment and dynamic regulations of dopamine receptors** *eLife* **13**:RP98358 <https://doi.org/10.7554/elife.98358.1> | Google Scholar
55. Aso Y., Siwanowicz I., Bräcker L., Ito K., Kitamoto T., Tanimoto H (2010) **Specific Dopaminergic Neurons for the Formation of Labile Aversive Memory** *Curr Biol* **20**:1445–1451 <https://doi.org/10.1016/j.cub.2010.06.048> | Google Scholar
56. Liu C., Plaçais P.-Y., Yamagata N., Pfeiffer B.D., Aso Y., Friedrich A.B., Siwanowicz I., Rubin G.M., Preat T., Tanimoto H (2012) **A subset of dopamine neurons signals reward for odour memory in *Drosophila*** *Nature* **488**:512–516 <https://doi.org/10.1038/nature11304> | Google Scholar
57. Aso Y., Rubin G.M (2016) **Dopaminergic neurons write and update memories with cell-type-specific rules** *eLife* **5**:e16135 <https://doi.org/10.7554/elife.16135> | Google Scholar
58. Yamagata N., Ichinose T., Aso Y., Plaçais P.-Y., Friedrich A.B., Sima R.J., Preat T., Rubin G.M., Tanimoto H (2015) **Distinct dopamine neurons mediate reward signals for short- and long-term memories** *Proc National Acad Sci* **112**:578–583 <https://doi.org/10.1073/pnas.1421930112> | Google Scholar
59. Aso Y., Herb A., Ogueta M., Siwanowicz I., Templier T., Friedrich A.B., Ito K., Scholz H., Tanimoto H (2012) **Three Dopamine Pathways Induce Aversive Odor Memories with Different Stability** *PLoS Genet* **8**:e1002768 <https://doi.org/10.1371/journal.pgen.1002768> | Google Scholar

60. Burke C.J., Huetteroth W., Oswald D., Perisse E., Krashes M.J., Das G., Gohl D., Silies M., Certel S., Waddell S (2012) **Layered reward signalling through octopamine and dopamine in *Drosophila*** *Nature* **492**:433–437 <https://doi.org/10.1038/nature11614> | Google Scholar
61. Baltruschat L., Prisco L., Ranft P., Lauritzen J.S., Fiala A., Bock D.D., Tavosan G (2021) **Circuit reorganization in the *Drosophila* mushroom body calyx accompanies memory consolidation** *Cell Reports* **34**:108871 <https://doi.org/10.1016/j.celrep.2021.108871> | Google Scholar
62. Tirian L., Dickson B.J (2017) **The VT GAL4, LexA, and split-GAL4 driver line collections for targeted expression in the *Drosophila* nervous system** *bioRxiv* :198648 <https://doi.org/10.1101/198648> | Google Scholar
63. Stocker R.F., Heimbeck G., Gendre N. (1997) **Neuroblast ablation in *Drosophila* P[GAL4] lines reveals origins of olfactory interneurons** *J. Neurobiol* **32**:443–456 [https://doi.org/10.1002/\(sici\)1097-4695\(199705\)32:5<443::aid-neu1>3.0.co;2-5](https://doi.org/10.1002/(sici)1097-4695(199705)32:5<443::aid-neu1>3.0.co;2-5) | Google Scholar
64. Pfeiffer B.D., Jenett A., Hammonds A.S., Ngo T.-T.B., Misra S., Murphy C., Scully A., Carlson J.W., Wan K.H., Lavery T.R., et al. (2008) **Tools for neuroanatomy and neurogenetics in *Drosophila*** *Proc. Natl. Acad. Sci* **105**:9715–9720 <https://doi.org/10.1073/pnas.0803697105> | Google Scholar
65. Ke M.-T., Nakai Y., Fujimoto S., Takayama R., Yoshida S., Kitajima T.S., Sato M., Imai T (2016) **Super-Resolution Mapping of Neuronal Circuitry With an Index-Optimized Clearing Agent** *Cell Reports* **14**:2718–2732 <https://doi.org/10.1016/j.celrep.2016.02.057> | Google Scholar
66. Sage D., Donati L., Soulez F., Fortun D., Schmit G., Seitz A., Guiet R., Vonesch C., Unser M (2017) **DeconvolutionLab2: An open-source software for deconvolution microscopy** *Methods* **115**:28–41 <https://doi.org/10.1016/j.ymeth.2016.12.015> | Google Scholar
67. Haase R., Royer L.A., Steinbach P., Schmidt D., Dibrov A., Schmidt U., Weigert M., Maghelli N., Tomancak P., Jug F., et al. (2020) **CLIJ: GPU-accelerated image processing for everyone** *Nat. Methods* **17**:5–6 <https://doi.org/10.1038/s41592-019-0650-1> | Google Scholar

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Reviewer #1 (Public review):

Summary:

The study by Wu et al. uses endogenous bruchpilot expression in a cell-type-specific manner to assess synaptic heterogeneity in adult *Drosophila melanogaster* mushroom body output neurons. The authors performed genomic on locus tagging of the presynaptic scaffold protein bruchpilot (BRP) with one part of splitGFP (GFP11) using the CRISPR/Cas9 methodology and co-expressed the other part of splitGFP (GFP1-10) using the GAL4/UAS system. Upon expression of both parts of splitGFP, fluorescent GFP is assembled at the N-terminus of BRP, exactly where BRP is endogenously expressed in active zones. For manageable analysis, a high-throughput pipeline was developed. This analysis evaluated parameters like location of BRP clusters, volume of clusters, and cluster intensity as a direct measure of the relative amount of BRP expression levels on site, using publicly available 3D analysis tools that are integrated in Fiji. Analysis was conducted for different mushroom body cell types in different mushroom body lobes using various specific GAL4 drivers. To test this new method of synapse assessment, Wu et al. performed an associative learning experiment in which an odor was paired with an aversive stimulus and found that, in a specific time frame after conditioning, the new analysis solidly revealed changes in BRP levels at specific synapses that are associated with aversive learning.

Strengths:

Expression of splitGFP bound to BRP enables intensity analysis of BRP expression levels as exactly one GFP molecule is expressed per BRP. This is a great tool for synapse assessment. This tool can be widely used for any synapse as long as driver lines are available to co-express the other part of splitGFP in a cell-type-specific manner. As neuropils and thus the BRP label can be extremely dense, the analysis pipeline developed here is very useful and important. The authors have chosen an exceptionally dense neuropil - the mushroom bodies - for their analysis and convincingly show that BRP assessment can be achieved with such densely packed active zones. The result that BRP levels change upon associative learning in an experiment with odor presentation paired with punishment is likewise convincing, and strongly suggests that the tool and pipeline developed here can be used in an *in vivo* context.

Weaknesses:

Although BRP is an important scaffold protein and its expression levels were associated with function and plasticity, I am still somewhat reluctant to accept that synapse structure profiling can be inferred from only assessing BRP expression levels and BRP cluster volume. Also, is it guaranteed that synaptic plasticity is not impaired by the large GFP fluorophore? Could the GFP10 construct that is tagged to BRP in all BRP-expressing cells, independent of GAL4, possibly hamper neuronal function? Is it certain that only active zones are labeled? I do see that plastic changes are made visible in this study after an associative learning experiment with BRP intensity and cluster volume as read-out, but I would be reassured by direct measurement of synaptic plasticity with splitGFP directly connected to BRP, maybe at a different synapse that is more accessible.

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Reviewer #2 (Public review):

Summary:

The authors developed a cell-type specific fluorescence-tagging approach using a CRISPR/Cas9 induced split-GFP reconstitution system to visualize endogenous Bruchpilot (BRP) clusters as presynaptic active zones (AZ) in specific cell types of the mushroom body (MB) in the adult *Drosophila* brain. This AZ profiling approach was implemented in a high-throughput quantification process, allowing for the comparison of synapse profiles within single cells, cell types, MB compartments, and between different individuals. The aim is to analyse in more detail neuronal connectivity and circuits in this centre of associative learning. These are notoriously difficult to investigate due to the density of cells and structures within a cell. The authors detect and characterize cell-type-specific differences in BRP-dependent profiling of presynapses in different compartments of the MB, while intracellular AZ distribution was found to be stereotyped. Next to the descriptive part characterizing various AZ profiles in the MB, the authors apply an associative learning assay and detect consequent AZ re-organisation.

Strengths:

The strength of this study lies in the outstanding resolution of synapse profiling in the extremely dense compartments of the MB. This detailed analysis will be the entry point for many future analyses of synapse diversity in connection with functional specificity to uncover the molecular mechanisms underlying learning and memory formation and neuronal network logics. Therefore, this approach is of high importance for the scientific community and a valuable tool to investigate and correlate AZ architecture and synapse function in the CNS.

Weaknesses:

The results and conclusions presented in this study are, in many aspects, well-supported by the data presented. To further support the key findings of the manuscript, additional controls, comments, and possibly broader functional analysis would be helpful. In particular:

(1) All experiments in the study are based on split-GFP lines (BRP:GFP11 and UAS-GFP1-10). The Materials and Methods section does not contain any cloning strategy (gRNA, primer, PCR/sequencing validation, exact position of tag insertion, etc.) and only refers to a bioRxiv publication. It might be helpful to add a Materials and Methods section (at least for the BRP:GFP11 line). Additionally, as this is an on locus insertion the in BRP-ORF, it needs a general validation of this line, including controls (Western Blot and correlative antibody

staining against BRP) showing that overall BRP expression is not compromised due to the GFP insertion and localizes as BRP in wild type flies, that flies are viable, have no defects in locomotion and learning and memory formation and MB morphology is not affected compared to wild type animals.

(2) Several aspects of image acquisition and high-throughput quantification data analysis would benefit from a more detailed clarification.

a) For BRP cluster segmentation it is stated in the Materials and Methods state, that intensity threshold and noise tolerance were "set" - this setting has a large effect on the quantification, and it should be specified and setting criteria named and justified (if set manually (how and why) or automatically (to what)). Additionally, if Python was used for "Nearest Neighbor" analysis, the code should be made available within this manuscript; otherwise, it is difficult to judge the quality of this quantification step.

b) To better evaluate the quality of both the imaging analysis and image presentation, it would be important to state, if presented and analysed images are deconvolved and if so, at least one proof of principle example of a comparison of original and deconvoluted file should be shown and quantified to show the impact of deconvolution on the output quality as this is central to this study.

(3) The major part of this study focuses on the description and comparison of the divergent synapse parameters across cell-types in MB compartments, which is highly relevant and interesting. Yet it would be very interesting to connect this new method with functional aspects of the heterogeneous synapses. This is done in Figure 7 with an associative learning approach, which is, in part, not trivial to follow for the reader and would profit from a more comprehensive analysis.

a) It would be important for the understanding and validation of the learning induced changes, if not (only) a ratio (of AZ density/local intensity) would be presented, but both values on their own, especially to allow a comparison to the quoted, previous AZ remodelling analysis quantifying BRP intensities (ref. 17, 18). It should be elucidated in more detail why only the ratio was presented here.

b) The reason why a single instead of a dual odour conditioning was performed could be clarified and discussed (would that have the same effects?).

c) Additionally, "controls" for the unpaired values - that is, in flies receiving neither shock nor odour - it would help to evaluate the unpaired control values in the different MB compartments.

d) The temporal resolution of the effect is very interesting (Figure 7D), and at more time points, especially between 90 and 270 min, this might raise interesting results.

e) Additionally, it would be very interesting and rewarding to have at least one additional assay, relating structure and function, e.g. on a molecular level by a correlative analysis of BRP and synaptic vesicles (by staining or co-expression of SV-protein markers) or calcium activity imaging or on a functional level by additional learning assays

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Reviewer #3 (Public review):

Summary:

The authors develop a tool for marking presynaptic active zones in *Drosophila* brains, dependent on the GAL4 construct used to express a fragment of GFP, which will incorporate

with a genome-engineered partial GFP attached to the active zone protein bruchpilot - signal will be specific to the GAL4-expressing neuronal compartment. They then use various GAL4s to examine innervation onto the mushroom bodies to dissect compartment-specific differences in the size and intensity of active zones. After a description of these differences, they induce learning in flies with classic odour/electric shock pairing and observe changes after conditioning that are specific to the paired conditioning/learning paradigm.

Strengths:

The imaging and analysis appear strong. The tool is novel and exciting.

Weaknesses:

I feel that the tool could do with a little more characterisation. It is assumed that the puncta observed are AZs with no further definition or characterisation.

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